Exercise JA

Q1. For every integer n, let a_n and b_n be real numbers. Let function f : IR → IR be given by

$f(x) = \begin{cases} a_n + \sin \pi x, & for\, x \in [2n, 2n+1] \\ b_n + \cos \pi x, & -for x \in (2n-1, 2n) \end{cases}$, for all integers n.

If f is continuous, then which of the following hold(s) for all n ?

(A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

Ans. B, D

Sol. For f to be continuous :

$f(2n^-) = f(2n^+)$

$\Rightarrow \quad b_n + \cos 2n\pi = a_n + \sin 2n\pi$

$\Rightarrow \quad b_n + 1 = a_n$

$\Rightarrow \quad a_n - b_n = 1$

($\therefore$ B is correct)

Also $f(x) = \begin{cases} b_n + \cos \pi x & (2n-1, 2n) \\ a_n + \sin \pi x & [2n, 2n+1] \\ b_{n+1} + \cos \pi x & (2n+1, 2n+2) \\ a_n + \sin \pi x & [2n+2, 2n+3] \end{cases}$

Again $f((2n+1)-) = f((2n+1)^+)$

$\Rightarrow \quad a_n = b_{n+1} - 1$

$\Rightarrow \quad a_n - b_{n+1} = -1$

$\Rightarrow \quad a_{n-1} - b_n = -1$ ($\therefore$ D is correct)

Q2. Let f(x) = $\begin{cases} x^2\left|\cos \frac{\pi}{x}\right|, & x \neq 0 \\ 0, & x = 4 \end{cases}$ x in R , then f is :

(A) differentiable both at x = 0 and at x = 2

(B) differentiable at x = 0 but not differentiable at x = 2

(C) not differentiable at x = 0 but differentiable at x = 2

(D) differentiable neither at x = 2 nor at x = 0

Ans. B

Sol. At x = 0

$$R.H.D = \lim_{h \to 0} \frac{(0+h) - (0)}{h} = \lim_{h \to 0} \frac{h^2\left|cos \frac{\pi}{h}\right| - 0}{h}$$

$= \lim_{h\to 0} h\left|cos\frac{\pi}{h}\right| = 0\times cos(\infty) = 0$ finite = 0

$$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{h^2cos\left(\frac{\pi}{-h}\right)-0}{-h} = \lim_{h\to 0} -hcos\left(\frac{\pi}{h}\right)$$

$= 0$

$\because$ LHD = RHD at x = 0

$\Rightarrow$ f(x) is differentiable at x = 0

At x = 2

$$RHD = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$$

$$= \lim_{h\to 0}\frac{(2+h)^2.\,cos\left(\frac{\pi}{2+h}\right)-0}{h}$$

$$= 4\lim_{h\to 0}\frac{cos\left(\frac{\pi}{2+h}\right)}{h}$$

$$= -4\lim_{h\to 0}\frac{sin\left(\frac{\pi}{2+h}\right).\left(-\frac{\pi}{(2+h)^2}\right)}{1} = \pi$$

$$LHD = \lim_{h\to 0}\frac{f(2-h)-f(2)}{-h}$$

$$= \lim_{h\to 0}\frac{(2-h)^2\left(cos-\left(\frac{\pi}{2-h}\right)\right)}{-h}$$

$$= \lim_{h\to 0}\frac{4\left(sin\frac{\pi}{(2-h)}\right)\left(\frac{\pi}{(2-h)^2}\right)}{-1} = -\pi$$

LHD $\neq$ RHD at x = 2

$\therefore$ Not differentiable at x = 2

Q3. Let $f_1 : R \to R$, $f_2 : [0, \infty) \to R$, $f_3 : R \to R$ and $f_4 : R \to [0,\infty)$ be defined by

$f_1(x) = \begin{cases} |x| & fi\ x < 0, \\ e^x & if\, x \geq 0; \end{cases}$ $f_2(x) = x^2$; $f_3(x) = \begin{cases} \sin x & fi\ x < 0, \\ x & if\, x \geq 0; \end{cases}$

and $f_4(x) = \begin{cases} f_2(f_1(x)) & fi\ x < 0, \\ f_2(f_1(x)) - 1 & if\, x \geq 0; \end{cases}$

List I	List II
P. f_4 is	1. onto but not one-one
Q. f_3 is	2. neither continuous nor one-one
R. $f_2 O f_1$ is	3. differentiable but not one-one
S. f_2 is	4. continuous and one-one

(A)
P	Q	R	S
3	1	4	2

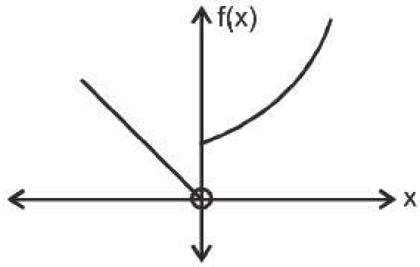
(B) 1 3 4 2 (C) 3 1 2 4 (D) 1 3 2 4

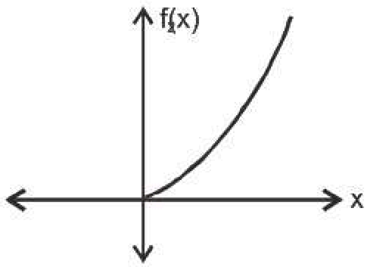
Ans. D

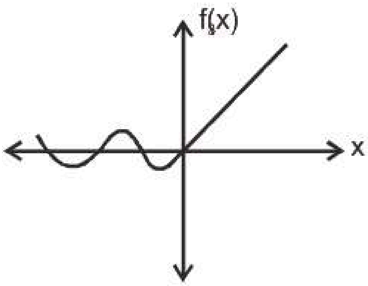
Sol.

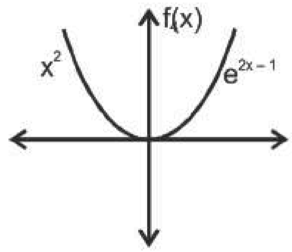
$$f_2(f_1(x)) = (f_1(x))^2 - \left[\begin{array}{ll} x^2 & x<0 \\ e^{2x} & x \geq 0 \end{array}\right.$$

$$f_4(x) = \left[\begin{array}{ll} x^2 & x<0 \\ e^{2x-1} & x \geq 0 \end{array}\right.$$









$f_4(x)$ is many-one onto, continuous and non-derivable

$f_3(x)$ is many-one, into, continuous and derivable

$f_2(x)$ is one-one, into, differentiable

Q4. Let f : R→ R and g : R → R be respectively given by f(x) = |x| +1 and g(x) = x^2 + 1. Define h : R → R by

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & if\, x \leq 0, \\ min\{f(x)g(x)\} & if\, x > 0 \end{cases}$$

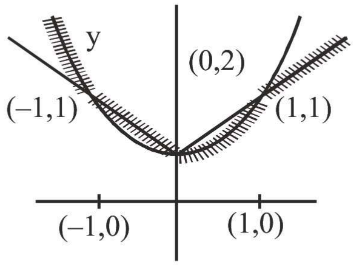
The number of points at which h(x) is not differentiable is :

Ans. 3

Sol.

$$f(x) = |x| + 1 = \begin{cases} x+1 & x \geq 0 \\ -x+1 & x < 0 \end{cases}$$

$$g(x) = x^2 = 1$$



Number of Non-differential points 3

Q5. Let g : R → R be a differentiable function with g (0) = 0, g'(0) = 0 and g'(1)≠ 0.

Let f (x) = $\begin{cases} \frac{x}{|x|}g, & x \neq 0 \\ 0, & x = 0 \end{cases}$ and h (x) = $e^{|x|}$ for all x ∈ R. Let (f o h) (x) denote f (h (x)) and

(h o f) (x) denote h (f(x)). Then which of the following is (are) true ?

(A) f is differentiable at x = 0 (B) h is differentiable at x = 0

(C) f o h is differentiable at x = 0 (D) h o f is differentiable at x = 0

Ans. A, D

Sol. $g(0) = 0, \quad g'(0) = 0 \quad g'(1) \neq 0$

$$f(x)\begin{cases} g(x) & ; \quad x > 0 \\ -g(x) & ; \quad x < 0 \\ 0 & ; \quad 0 \end{cases} \qquad h(x) = e^{|x|}$$

$$f(h(x)) = g\left(e^{|x|}\right), h(f(x)) = e^{|g(x)|}$$

$$R(f'(0)) = \lim_{x\to 10} \frac{g(x) - g(0)}{x - 0} = g'(0) = 0$$

$$L(f'(0)) = \lim_{x\to 10} \frac{-g(x) - g(0)}{x - 0} = g'(0) = 0$$

$$R(h'(0)) = 1 \& L(h'(0)) = -1$$

So h(x) is non derivable. at x = 0

$$\text{Now} \lim_{x\to 0} \frac{f(h(x)) - f(h(0))}{x} = \lim_{x\to 0} \frac{g(e^{|x|}) - g(1)}{x}$$

$$R(f'(0)) = \lim_{x\to 0} \frac{f(h(x)) - f(h(0))}{x} = \lim_{x\to 0} \frac{g(e^{|x|}) - g(1)}{e^x - 1} \frac{e^x - 1}{x} = g'(1)$$

$$L\left(f'\left(h(x)\right)\right) = -g'(1)$$

Hence is non derivable at x = 0

Since x = 0 is repeated root of g(x)

So is differetiable at x = 0

hence (A), (D)

Q6. Let a, b $\in$ R and f : R $\to$ R be defined by $f(x) = a\cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$. Then f is

(A) differentiable at x = 0 if a = 0 and b = 1

(B) differentiable at x = 1 if a = 1 and b = 0

(C) NOT differentaible at x=0 at a=1 and b=0

(D) NOT differentiable at x=1 if a=1 and b=1

Ans. A, B

Sol. $f(x) = a\cos(x^3 - x) + bx\sin(x^3 + x) \quad \forall x \in R$

Which is composition and sum of differentiable functions,

therefore f(x) is always continuous and differentiable.

Q7. Let $\left[-\frac{1}{2},2\right] \to \square$ and $g\left[-\frac{1}{2},2\right] \to \square$ be functions defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|\, f(x) + |4x - 7|\, f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \square$. Then

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2},2\right]$

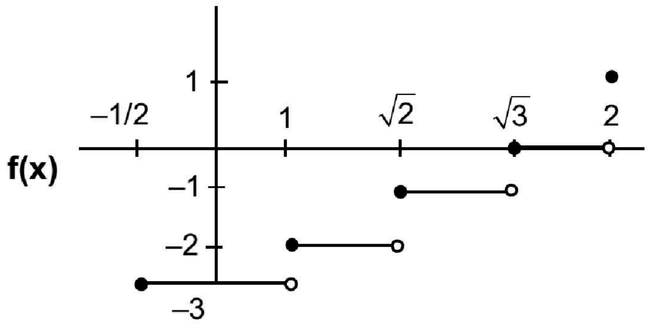
(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2},2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2},2\right)$

(D) g is NOT differentiable exactly at five point in $\left(-\frac{1}{2},2\right)$

Ans. B, C

Sol.



Clearly from the graph f is discontinuous at four points.

$g(x) = f(x)(|x| + |4x - 7|)$

f(x) is non-differentiable at $x = 1, \sqrt{2}, \sqrt{3}$ & $|x| + |4x - 7|$ is non-differentiable at $x = 0, \frac{7}{4}$

But $f(x) = 0\ \forall x \in \left[\sqrt{3}, 2\right]$

Hence g(x) is non-differentiable at $x = 0, 1, \sqrt{2}, \sqrt{3}$

Q8. Let [x] be the greatest integer less than or equal to x. Then, at which of the following point(s) the function
$f(x) = x\cos(\pi([x] + x))$ is discontinuous?

(A) $x = 2$ (B) $x = 1$ (C) $x = -1$ (D) $x = 0$

Ans. A, B, C

Sol. $f(x) = \cos(\pi x + \pi[x])$

f(x) is continuous at $x = 0$

f(x) is disc. at $x = 2, 1, -1$

$$f(x) = \begin{cases} x\cos \pi x & [x] = \text{even} \\ -\cos \pi x & [x] = \text{odd} \end{cases}$$

Now check at integer

f(x) is continuous at x = 0

f(x) is disc. at x = 2, 1, –1

Q9. Let $f_1 : \square \to \square$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \square$, $f_3 \left(-1, e^{\frac{\pi}{2}} - 2\right) \to \square$ and $f_4 : \square \to \square$ defined by

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{e^{-x^2}}}\right)$

(ii) $\begin{cases} \frac{|\sin x|}{\tan^{-1} x} & if\, x \neq 0 \\ 1 & x = 0 \end{cases}$, where the inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, where, for $\in \square$ [t] denotes the greatest integer less than or equal to t,

(iv) $f4\,(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & if\ \neq 0 \\ 0 & if\ x = 0 \end{cases}$

LIST-I	LIST-II
P. The function f_1 is	1. NOT continuous at x = 0
Q. The function f_2 is	2. continuous at x = 0 and NOT differentiable at x = 0
R. The function f_3 is	3. differentiable at x = 0 and its derivative is NOT continuous at x = 0
S. The function f_4 is	4. differentiable at x = 0 and its derivative is continuous at x = 0

The correct option is:

(A) $P \to 2; Q \to 3; R \to 1; S \to 4$ (B) $P \to 4; Q \to 1; R \to 2; S \to 3$

(C) $P \to 4; Q \to 2; R \to 1; S \to 3$ (D) $P \to 2; Q \to 1; R \to 4; S \to 3$

Ans. D

Sol. (i) $f'_1(0) = \lim\limits_{h \to 0} \frac{sin\sqrt{1-e^{-h^2}} - 0}{h} = \lim\limits_{h \to 0} \frac{sin\sqrt{1-e^{-h^2}}}{\sqrt{1-e^{-h^2}}} \times \sqrt{\frac{1-e^{-h^2}}{h^2}} \times \frac{|h|}{h}$

$= 1 \times 1 \times \frac{|h|}{h} = 1 \times 1 \times \frac{|h|}{h}$

= limit does not exist.

$\Rightarrow$ for option (P), (2) is correct.

(ii) $\lim_{x\to0} f_2(x) = \lim_{x\to0} \frac{|sinx}{tan^{-1}x}$

$= \lim_{x\to0} \frac{|sinx|}{|x|} \times \frac{x}{tan^{-1}x} \times \frac{|x|}{x}$

$= \lim_{x\to0} 1\times1\times\frac{|x|}{x}$

= limit does not exist $\Rightarrow$ for option Q, (1) is correct.

(iii) $\lim_{x\to0} f_3(x) = \lim_{x\to0} [\sin(\log_e(x+2)\text{ tends to }2]$

now at x tends to zero (x + 2) tends to 2

$\Rightarrow$ $\log_e$ (x + 2) tends to ln2

which is less than 1

$0 < \lim_{x\to0} sin(log_e(x+2)) < sin1$

$\Rightarrow f_3(x) = \{0 \quad x \in [-1, e^{\frac{\pi}{2}}-2)$

$\Rightarrow f'{}_3(x) = 0\ \forall x \in (-1, e^{\frac{\pi}{2}}-2)$

$\Rightarrow f''{}_3(x) = 0\ \forall x \in (-1, e^{\frac{\pi}{2}}-2)$

Hence for (R), (4) is correct.

(iv) $\lim_{x\to0} f_4(x) = \lim_{x\to0}\left(x^2 sin\frac{1}{x}\right) = \lim_{x\to0} x^2\left(sin\frac{1}{x}\right) = 0$

$f'{}_4(0) = \lim_{x\to0} \frac{h^2 sin\left(\frac{1}{x}\right) - 0}{x} = \lim_{x\to0} h\, sin\left(\frac{1}{x}\right) = 0$

$f'{}_4(x) = -cos\frac{1}{x} + xsin\frac{1}{x}, x \neq 0$

$f''_4(0) = \frac{-cos\frac{1}{h} + h\, sin\frac{1}{h} - 0}{h}$ $\Rightarrow$ does not exist.

hence for (S), (3) is correct.

Q10. Let f : R→R be a function. We say that f has

PROPERTY 1 if $\lim_{h\to0} \frac{f(h)-f(0)}{\sqrt{|h|}}$ exists and is finite and

PROPERTY 2 if $\lim_{h\to0} \frac{f(h)-f(0)}{h^2}$ exists and is finite and

Then which of the following options is/are correct ?

(A) f(x) = $|x|$ has PROPERTY 1 (B) f(x) = $x|x|$ has PROPERTY 2

(C) f(x) = sin x has PROPERTY 2 (D) f(x) = $x^{2/3}$ has PROPERTY 1

Ans. A, D

Sol. Option (1) f (x) = |x| f(0) = 0

Property -1

$$\lim_{h\to 0}\frac{|h|}{\sqrt{|h|}}=\lim_{h\to 0}|h|^{\frac{1}{2}}=0 \quad \text{limit exists}$$

Option (2) f (x) = x|x| f (0) = 0

Property -2

$$\lim_{h\to 0}\frac{h|h|-0}{h^2}=\lim_{h\to 0}|\frac{|h|}{h}$$

LHL = –1

RHL = 1

LHL $\neq$ RHL

$\Rightarrow$ Does not exist

Option (3) f (x) = sinx   f (0) = 0

Property - 2

$$\lim_{h\to 0}\frac{sin\,h}{h^2}=\lim_{h\to 0}\left(\frac{sin\,h}{h}.\frac{1}{h}\right)$$

$\Rightarrow$ Does not exist

Option (4) $f(x)=x^{\frac{2}{3}}$

$(x^2)^{\frac{1}{3}}=|x|^{\frac{2}{3}}$ $f(0)=0$

Property -1

$$\lim_{h\to 0}\frac{|h|^{\frac{2}{3}}-0}{\sqrt{|h|}}=\lim_{h\to 0}|h|^{\frac{2}{3}-\frac{1}{2}}=\lim_{h\to 0}|h|^{\frac{1}{6}}=0 \qquad \text{limit exist}$$

Q11. Let the function f : R $\to$ R be defined by f (x) = $x^3 - x^2$ + (x – 1) sin x and let g : R $\to$ R be an arbitrary function. Let f g : R $\to$ R be the product function defined by (f g) (x) = f (x) g (x). Then which of the following statements is/are TRUE?

(A) If g is continuous at x =1, then f g is differentiable at x =1

(B) If f g is differentiable at x = 1, then g is continuous at x =1

(C) If g is differentiable at x =1, then f g is differentiable at x =1

(D) If f g is differentiable at x =1, then g is differentiable at x =1

Ans. A, C

Sol. f(x) = $x^3 - x^2$ + (x – 1) sin (x)
f (1) = 0
f is continous & differentiable
f'(1) = 1 + sin1

Let h (x) = f (x) g (x), h (1) = 0

Differentiability at x = 1

$h'1^{+} = \lim\limits_{h \to \infty} \dfrac{f(1+h)g(1+h) - 0}{h}$

= f' (1) g (1^+)

$h'(1^{-}) = \lim\limits_{h \to \infty} \dfrac{f(1-h)g(1-h) - 0}{-h}$

= f '(1) g(1^-)

(A) If g is continuous at x = 1

⇒ g (1^+) = g(1) = g (1^-)

⇒ h' (1^+) = h' (1^-)

(B) If h' (1^+) = h' (1^-)

⇒ g (1^+) = g (1^-)

but g(1) can be different

(C) If g is differentiable at x = 1

⇒ g is continuous at x = 1

from A we can say that C is true

(B) is false so (D) is also false

Q12. Let f : R → R and g : R → R be functions satisfying f(x + y) = f(x) + f(y) + f(x)f(y) and f(x) = xg(x) for all x, y ∈ R. If $\lim\limits_{x \to 0}$ g (x) = 1, then which of the following statements is/are TRUE?

(A) f is differentiable at every x ∈ R

(B) If g(0) = 1, then g is differentiable at every x ∈ R

(C) The derivative $f'(1)$ is equal to 1

(D) The derivative $f'(0)$ is equal to 1

Ans. A, B, D

Sol. since $f(\mathrm{x}) = \mathrm{xg}(\mathrm{x})$

$\lim\limits_{x \to 0} f(x) = \lim\limits_{x \to 0} xg(x)$

$\lim\limits_{x \to 0} f(x) = \left(\lim\limits_{x \to 0} x\right) \cdot \left(\lim\limits_{x \to 0} g(x)\right)$

$\lim\limits_{x \to 0} f(x) = 0 \times 1 = 0$ ....(1)

$f(\mathrm{x}+\mathrm{y}) = f(\mathrm{x}) + f(\mathrm{x}) + f(\mathrm{x}) f(\mathrm{y})$

Now we check continuity of $f(\mathrm{x})$

at x = a

$\lim\limits_{h \to 0} f(a+h) = f(a) + f(b) + f(a) + f(h)$

$\lim\limits_{x \to 0} (f(a) + f(h)(1 + f(a)))$

$\lim\limits_{h \to 0} f(a+h) = f(a)$

$\lim_{h\to 0} f(a+h) = f(a)$

$\therefore f(\mathrm{x})$ is continuous $\forall \; x \in \mathrm{R}$

$\lim_{x\to 0} f(x) = f(0) = 0 \quad \left(\lim_{x\to 0} f(x) = 0\right)$

$\therefore f(0) = 0$

and $\lim_{x\to 0} \frac{f'(x)}{1} = 1$

$\therefore f'(0) = 1$

Now

$f(\mathrm{x}+\mathrm{y}) = f(\mathrm{x}) + f(\mathrm{y}) + f(\mathrm{x})f(\mathrm{y})$

using partial derivative (w.r.t. y)

$f'(\mathrm{x}+\mathrm{y}) + f'(\mathrm{y}) + f(\mathrm{x}) + f'(\mathrm{y})$

put y = 0

$f'(\mathrm{x}) = f'(0) + f(\mathrm{x})f'(0)$

$f'(\mathrm{x}) = 1 + f(\mathrm{x})$

$\int \frac{f'(x)}{1+f(x)} dx = \int 1dx$

$\ell n|(1+f(x))| = x + C$

$f(0) = 0; \mathrm{c} = 0 \quad \therefore |1 + f(\mathrm{x})| = \mathrm{e}^{\mathrm{x}}$

$1 + f(\mathrm{x}) = \pm \mathrm{e}^{\mathrm{x}}$ *or* $f(\mathrm{x}) = \pm\mathrm{e}^{\mathrm{x}} - 1$

Now $f(0) = 0 \qquad \therefore f(\mathrm{x}) = \mathrm{e}^{\mathrm{x}} - 1$

$\therefore f(\mathrm{x}) = \mathrm{e}^{\mathrm{x}} - 1$

option (A) is correct

$and \quad f'(x) = e^x$

f' (0) = 1 option (D) is correct

$g(x) = \frac{f(x)}{x} = \begin{Bmatrix} \frac{e^x-1}{x} & ; & x \neq 0 \\ 1 & ; & x = 0 \end{Bmatrix}$

$g'(0+h) = \lim_{h\to 0} \frac{g(0+h) - g(0)}{h}$

$= \lim_{h\to 0} \frac{\frac{e^h-1}{h} - 1}{h} = \frac{1}{2}$

option B is correct

Q13. Let the functions f : (–1, 1) → R and g : (–1, 1) → (–1, 1) be defined by f(x) = |2x – 1| + |2x + 1| and g(x) = x – [x],
where [x] denotes the greatest integer less than or equal to x. Let f o g : (–1, 1) → R be the composite function defined by (f o g) (x) = f (g(x)). Suppose c is the number of points in the interval (–1, 1) at which f o g is NOT continuous, and suppose d is the number of points in the interval (–1, 1) at which f o g is NOT differentiable. Then the value of c + d is ———.

Ans. 4

Sol. f(x) = |2x –1| + |2x + 1| 　　　　g(x) = x – [x]

g(x) = [x] + {x} – [x]

$$g(x) = \{x\}$$

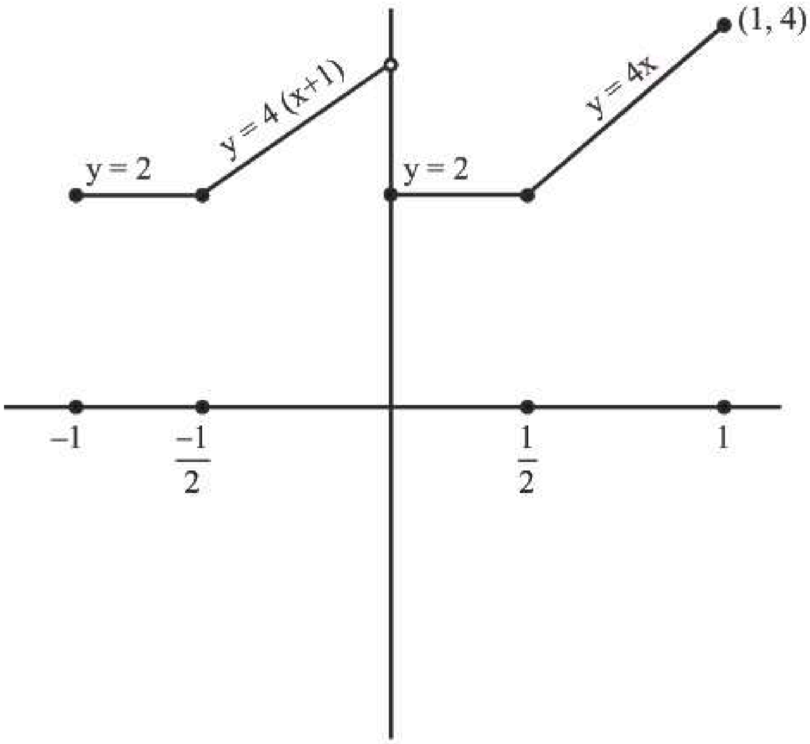
$$f(x) = \begin{cases} 4x & : x \ge \frac{1}{2} \\ 2 & : \frac{-1}{2} < x < \frac{1}{2} \\ -4x & : x \le \frac{-1}{2} \end{cases}$$

$$f\left(g(x)\right) = \begin{cases} 4\{x\} & : \{x\} \ge \frac{1}{2} \\ 2 & : \frac{-1}{2} < \{x\} < \frac{1}{2} \\ -4\{x\} & : \{x\} \le \frac{-1}{2} \end{cases}$$

$$fg((x)\ ) = \begin{cases} 4\{x\} & : \{x\} \ge \frac{1}{2} \\ 2 & : 0 \le \{x\} < \frac{1}{2} \end{cases}$$

$$fg(x)\ ) = \begin{cases} 4\{x\} & : x \in \left[\frac{1}{2}, 1\right) \cup \left[\frac{-1}{2}, 0\right) \\ 2 & : \left[0, \frac{1}{2}\right) \cup \left(\frac{-1}{2}, -1\right) \end{cases}$$

$$f\left(g(x)\right) = \begin{cases} 4x & : x \in \left[\frac{1}{2}, 1\right) \\ 4\left(x+1\right) & : x \in \left[\frac{-1}{2}, 0\right) \\ 2 & : x \in \left[0, \frac{1}{2}\right) \cup \left(\frac{-1}{2}, -1\right) \end{cases}$$



Not continuous at x = 0

Not differetiable at x = $\pm\frac{1}{2}, 0$

c = 1

d = 3

c + d = 1 + 3 = 4

Q14. Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

(A) There are infinitely many functions from S to T

(B) There are infinitely many strictly increasing functions from S to T

(C) The number of continuous functions from S to T is at most 120

(D) The number of continuous functions from S to T is at most 120

Ans. A, B, D

Sol. $S = (0,1) \cup (1,2) \cup (3,4)$

T = {0,1,2,3}

Number of functions

Each element of S have 4 choice

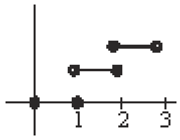
Let n be the number of element in set S.

Number of function = 4^n

Here $n \to \infty$

$\Rightarrow$ Option (A) is correct.

Option (B) is incorrect (obvious)



For continuous function

Each interval will have 4 choices

$\Rightarrow$ Number of continuous functions

$= 4\times4\times4 = 64$

$\Rightarrow$ Option (C) is correct.

(D) Every continuous function is piecewise constant functions

$\Rightarrow$ Differentiable.

Option (D) is correct.

Q15. Let $f:(0,1)\to R$ be the function defined as $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, where [x] denotes the greatest integer less than or equal to x. Then which of the following statements is (are) true?

(A) The function f is discontinuous exactly at one point in (0, 1)

(B) There is exactly one point in (0, 1) at which the function f is continuous but NOT differentiable

(C) The function f is NOT differentiable at more than three points in (0, 1)

(D) The minimum value of the function f is $-\frac{1}{512}$

Ans. A, B

Sol. $f:(0,1)\to R$

$f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right) \quad \Rightarrow \text{Criticalpoint} = \frac{1}{4},\frac{1}{2},\frac{3}{4}$

Discontinuity at $x = \frac{3}{4}$

Continuous and differentiable at $x = \frac{1}{4}$

Continuous but non-differentiable at $x = \frac{1}{2}$

$LHD\left(at\ x=\frac{1}{4}\right)$ $\qquad$ $RHD\left(at\ x=\frac{1}{4}\right)$

$\lim\limits_{h\to0^+}\frac{0-0}{-h}=0$ $\qquad$ $\lim\limits_{h\to0^+}\frac{h^2\left(-\frac{1}{2}+h\right)}{h}=0$

$LHD\left(at\ x=\frac{1}{2}\right)$ $\qquad$ $RHD\left(at\ x=\frac{1}{2}\right)$

$$\lim_{h\to 0^+}\frac{\left(\frac{1}{4}-h\right)^2(-h)-0}{-h}-\frac{1}{16} \qquad \lim_{h\to 0^+}\frac{2\left(\frac{1}{4}+h\right)^2h-0}{h}-\frac{1}{8}$$

Minimum -ve value will exist between $\frac{1}{4}$ and $\frac{1}{2}$

$$f(x)=\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right) \qquad \frac{1}{4}\le x\le\frac{1}{2}$$

$$f'(x)=\left(x-\frac{1}{4}\right)\left(3x-\frac{5}{4}\right) \quad \Rightarrow \quad \text{minima at } x=\frac{5}{12}$$

$$f\left(\frac{5}{12}\right)=\frac{1}{36}\times\frac{-1}{12} \quad =\frac{-1}{432}$$

Q16. Let f : R → R and g : R → R be functions defined by

$$f(x)=\begin{cases} x|x|\sin\left(\frac{1}{x}\right) & , x\ne 0 \\ 0, & x=0,\end{cases} \text{ and } g(x)=\begin{cases} 1-2x, & 0\le x\le\frac{1}{2}, \\ 0, & otherwise,\end{cases}.$$

Let a,b,c,d ∈ R. Define the function h : R → R by

$$h(x)=a\,f(x)\ +b\left(g(x)+g\left(\frac{1}{2}-x\right)\right)+c\,(x-g(x))+dg(x), x\in R.$$

Match each entry in List-I to the correct entery in List -II.

List-I	**List-II**
(P) If a = 0, b = 1, c = 0, and d = 0, then	(1) h is one-one
(Q) If a = 1, b = 0, c = 0, and d = 0, then	(2) h is onto
(R) If a = 0, b = 0, c = 1, and d = 0, then	(3) h is differentiable on R
(S) If a = 0, b = 0, c = 0, and d = 1, then	(4) the range of h is [0, 1]
	(5) the range of h is {0, 1}

The correct option is :

(A) (P) → (4); (Q) → (3); (R) → (1); (S) → (2)

(B) (P) → (5); (Q) → (2); (R) → (4); (S) → (3)

(C) (P) → (5); (Q) → (3); (R) → (2); (S) → (4)

(D) (P) → (4); (Q) → (2); (R) → (1); (S) → (3)

Ans. 66619b4af39b1c45c6dd7d46

Sol. (P) $h(x)=g(x)+g\left(\frac{1}{2}-x\right)$

$$g(x)=\begin{bmatrix} 1-2x & ; & 0\le x\le\frac{1}{2} \\ 0 & ; & otherwise\end{bmatrix}$$

$$g\left(\frac{1}{2}-x\right)=\begin{bmatrix} 2x & : & 0\le x<\frac{1}{2} \\ 0 & : & otherwise\end{bmatrix}$$

$$h(x)=\begin{cases}1 & : \quad 0\le x<\frac{1}{2}\\ 0 & : \quad otherwise\end{cases}$$

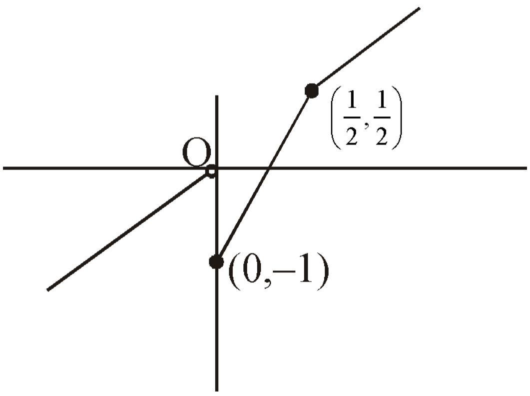
hence, correct option is (5).

(Q) $h(x)=\begin{cases}x|x|\sin\frac{1}{x}; & x\neq 0\\ 0; & otherwise\end{cases}$

$$h(x)=\begin{cases}x^2\sin\frac{1}{x} & : \quad x>0\\ -x^2\sin\frac{1}{x} & : \quad x<0\\ 0 & : \quad x=0\end{cases}$$

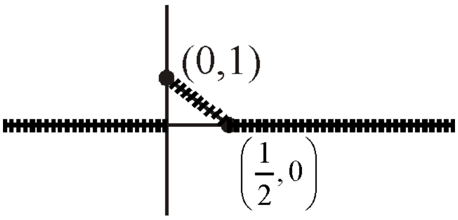
Clearly, function is differentiable everywhere. Hence correct option is (3).

(R) $h(x)=\begin{cases}3x-1 & : \quad 0\le x\le\frac{1}{2}\\ x & : \quad otherwise\end{cases}$



Clearly onto. Hence, correct option is (2).

(S) $g(x)=\begin{cases}1-2x & : \quad 0\le x\le\frac{1}{2}\\ 0 & : \quad otherwise\end{cases}$



Clearly, range is [0,1].
Hence, correct option is (4).
Hence, overall answer is (C).

Exercise C1

Q1. Which statement is false :

(A) f(x) = $\lim\limits_{n\to\infty} \dfrac{1}{1+n\sin^2\pi x}$ is continuous at x = 1.

(B) The function f(x) = $2^{-2^{1/(1-x)}}$ if x $\neq$ 1 & f (1) = 1 is not continuous at x = 1.

(C) There exists a continuous function f: [0, 1] onto [0, 10], but there exists no continuous function g : [0, 1] onto (0, 10).

(D) If f (x) is continuous in [0 , 1] & f(x) = 1 for all rational numbers in [0 , 1] then f $(1/\sqrt{2})$ equal to 1.

Ans. A

Sol. -

Q2. Given f(x) = $\dfrac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)sgn(\sin x)}$ for x $\neq$ 0

= 0 for x = 0

where {x} is the fractional part function; [x] is the step up function and sgn(x) is the signum function of x then, f(x)

(A) is continuous at x = 0 (B) is discontinuous at x = 0

(C) has a removable discontinuity at x = 0

(D) has an irremovable discontinuity at x = 0

Ans. A

Sol. R.H.L. = 0 Continuous at x = 0
L.H.L.= 0
f(0) =0

Q3. Consider f(x) = $\left[\begin{array}{l} x[x]^2\log_{1+x}2 for -1<x<0 \\ \dfrac{ln\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} for 0<x<1 \end{array}\right.$

where [ ] & {} are the greatest integer function & fractional part function respectively, then

(A) f(0) = $\ln^2$ ⇒ f is continuous at x = 0 (B) f(0) = 2 ⇒ f is continuous at x = 0

(C) f(0) = e^2 ⇒ f is continuous at x = 0

(D) f has an irremovable discontinuity at x = 0

Ans. D

Sol. at x = 0

$$R.H.L = \frac{ln\left(e^{h^2} + 2\sqrt{\{h\}}\right.}{tan\sqrt{h}}$$

$$= \frac{ln\left(e^{h^2} + 2\sqrt{\{h\}}\right)}{\sqrt{h}}$$

$$= \frac{ln\left(1 + e^{h^2} + 2\sqrt{h} - 1\right)\left(e^{h^2} + 2\sqrt{h} - 1\right)}{\sqrt{h}\left(e^{h^2} + 2\sqrt{h} - 1\right)}$$

$$= \frac{e^{h^2} + 2\sqrt{h} - 1}{\sqrt{h}}$$

$$= \frac{e^{h^2} - 1}{\sqrt{h}} + 2$$

$$= h^{\frac{3}{2}}\left(\frac{e^{h^2} - 1}{h}\right) + 2 = 2$$

$$= L.H.L. = h[-h]^2\, log_{1-h}^{2}$$

$$= \frac{-h\times(1)ln2\times(-1)}{ln(1-h)} = -ln2$$ Correct option.

Q4. Consider $f(x) = \frac{\sqrt{1+x} - \sqrt{1-x}}{\{x\}}$, $x \neq 0$;

$g(x) = \cos 2x,\ -\frac{\pi}{4} < x < 0,$

$$h(x) - \quad f(x) = \begin{cases} -x, -\frac{\pi}{2} & x \le \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \le 0 \\ x - 1, & 0 < x \le 1 \\ ln\, x & x > 1 \end{cases}$$

then, which of the following holds good.

where {x} denotes fractional part function.

(A) 'h' is continuous at x = 0 (B) 'h' is discontinuous at x = 0

(C) f(g(x)) is an even function (D) f(x) is an even function

Ans. A

Sol. at x= 0

h(0) = 1

R.H.L. = f(0 + h)

$$= \frac{\sqrt{1+h}-\sqrt{1-h}}{\{h\}}$$

$$= \frac{\sqrt{1+h}-\sqrt{1-h}}{h}$$

$$\frac{1+h-1+h}{h} \times \frac{1}{\sqrt{1+h}+\sqrt{1-h}} = 1$$

$$L.H.L = \frac{1}{\sqrt{2}} f(g(-h)$$

$$= \frac{1}{\sqrt{2}} \frac{\sqrt{1+cos2h}-\sqrt{1-cos2h}}{\{cos2h\}}$$

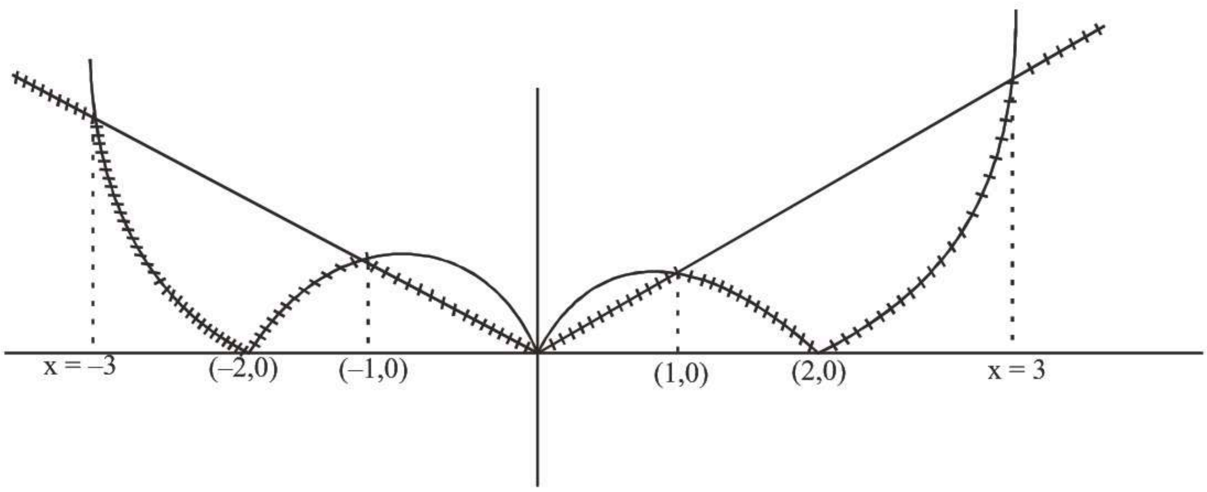
$$= \frac{1}{\sqrt{2}} \frac{2cos2h}{cos2h} \times \left(\frac{1}{\sqrt{1+cosh}+\sqrt{1+cos2h}}\right)$$

$$= \sqrt{2} \times \frac{2}{2\sqrt{2}} = 1$$ correct option (A)

g(x) = Min. $\{|x^2 - 2|x|, |x|\}$

$y = |x^2 - 2|x||$

$y = |x|$



hence it is clear g(x) is not differentiable at x = ± 2, ± 1, 0

Q5. Consider the function $f(x) = \lim_{n \to \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{n+1} - x^{2n}}$,where n $\in$ N

Statement-1: f (x) is discontinuous at x = 1. because

Statement-2: f (1) = 0.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Ans. B

Sol. $f(x) = \lim\limits_{n\to\infty} \dfrac{\sin\pi - (x^2)^{n\sin}(x-1)}{1 + x(x^2)^n - (x^2)^n}$

$$f(x) = \begin{cases} \dfrac{\sin\pi x}{1} & ;\ 0 < x^2 < \\[2ex] \dfrac{\sin\pi x - \sin(x-1)}{x} & ;\ x^2 = 1 \\[2ex] \dfrac{-\sin(x-1)}{x-1} & ;\ x^2 > 1 \end{cases}$$

at $x = 1$

$f(1) = 0$

$$R.H.L = \frac{-sin(1+h-1)}{(1+h-1)} = -sin\pi h = 0$$

$$L.H.L = \frac{sin(\pi - \pi h)}{1} = -sin\pi h = 0$$

Q6. Given $f(x) = \begin{cases} \log_a \left(a\left|[x] + [-x]\right|\right)^x \left(\dfrac{a^{\left(\frac{2}{\frac{[x]+[-x]}{|x|}}\right)} - 5}{3 + a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0\ ;\ a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ where [ ] represents the integral part function, then :

(A) f is continuous but not differentiable at $x = 0$ (B) f is cont. & diff. at $x = 0$

(C) the differentiability of 'f' at $x = 0$ depends on the value of a

(D) f is cont. & diff. at $x = 0$ and for $a = e$ only.

Ans. B

Sol. Obvious

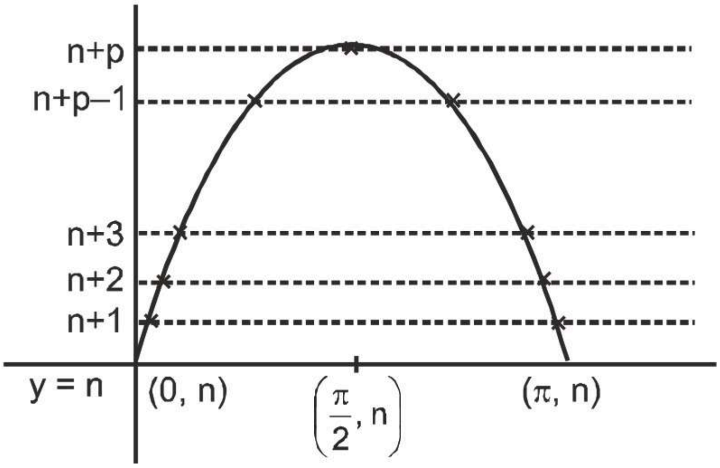
Q7. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ and p is a prime number. The number of points where $f(x)$ is not differentiable is
Here [x] denotes greatest integer function.

(A) $p - 1$ (B) $p + 1$ (C) $2p + 1$ (D) $2p - 1$.

Ans. B

Sol. $f(x) = [n + p\sin x]$, $x \in (0, \pi)$

graph of $y = n + p \sin x$



obviously

$f(x) = [n + p\sin x]$ is discontinous at points mark in above curve number of such points $\Rightarrow (p - 1) + 1 + p - 1 = 2p - 1$

Q8. Consider $f(x) = \left[\dfrac{2(\sin x - \sin^3 x) + |\sin x - \sin^3 x|}{2(\sin x - \sin^3) - |\sin x - \sin^3 x|}\right]$, $x \neq \dfrac{\pi}{2}$ for $x \in (0, \pi)$

$f(\pi/2) = 3$ where [ ] denotes the greatest integer function then,

(A) f is continuous & differentiable at $x = \pi/2$

(B) f is continuous but not differentiable at $x = \pi/2$

(C) f is neither continuous nor differentiable at $x = \pi/2$ (D) none of these

Ans. A

Sol. In the immediate neighborhood of $x = \dfrac{\pi}{2}$, $\sin x > \sin^3 x \Rightarrow |\sin x - \sin^3 x| = \sin x - \sin^3 x$

Hence for $x \neq \dfrac{\pi}{2}$, $f(x) = \left[\dfrac{2(\sin x - \sin^3 x) + \sin x - \sin^3 x}{2(\sin x - \sin^3 x) - \sin x + \sin^3 x}\right] = \dfrac{3\sin x - 3\sin z^3 x}{\sin x - \sin^3 x} = 3$

Hence f is continuous and diff. at $x = \dfrac{\pi}{2}$]

Q9. Let $f : [a, b] \to R$ be any function which is such that $f(x)$ is rational for irrational x and that $f(x)$ is irrational for rational x, then in [a, b]

(A) f is discontinuous everywhere in [a, b]

(B) f is continuous only at x = 0 and discontinuous everywhere

(C) f is continuous for all irrational x and discontinuous for rational x

(D) f is continuous for rational x and discontinuous for irrational x.

Ans. A

Sol. $f(x) = \begin{cases} rational & if\ x \in Q \in [a,b] \\ irrational & if\ x \in Q \in [a,b] \end{cases}$

$$\left.\begin{array}{l} f(c) = \text{irrational} \\ f(c+h) = \text{rational} \\ \lim\limits_{h\to 0} \text{through irrational} \end{array}\right] \Rightarrow \quad f \text{ is discontinuous for } x \in Q \text{ in } [a, b]$$

now let $r \in Q$

$$\left.\begin{array}{l} f(r) = \text{rational} \\ f(r+h) = \text{irrational} \\ \lim\limits_{h\to 0} \text{through rational} \end{array}\right] \Rightarrow \quad f \text{ is discontinuous for } x \in Q \text{ in } [a, b]$$

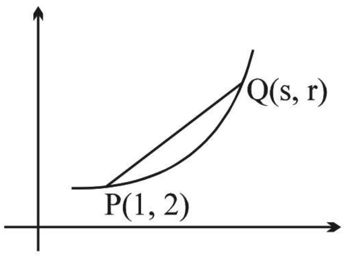
Hence f is discontinuous every where in [a, b].

Q10. The graph of function f contains the point P (1, 2) and Q(s, r). The equation of the secant line through P and Q is $y = \left(\frac{s^2+2x-3}{s-1}\right)x - 1 - s$. The value of $f'(1)$, is

(A) 2 (B) 3 (C) 4 (D) non existent

Ans. C

Sol.



I. By definition $f'(1)$ is the limit of the slope of the secant line when $s \to 1$.

Thus $f'(1) = \lim\limits_{s\to 1} \frac{s^2+2s-3}{s-1}$

$= \lim\limits_{s\to 1} \frac{(s-1)(s+3)}{s-1}$

$= \lim\limits_{s\to 1}(s+3) = 4 \quad \Rightarrow \quad$ (D)

II. By substituting x = s into the equation of the secant line, and cancelling by s – 1 again, we get

$y = s^2 + 2s - 1$. This is f (s), and its derivative is

$f'(s) = 2s + 2$, so $f'(1) = 4$.

Q11. If $f(x + y) = f(x) + f(y) + |x|y + xy^2$, $\forall$ x, y $\in$ R and $f'(0) = 0$, then

(A) f need not be differentiable at every non zero x (B) f is differentiable for all $x \in R$

(C) f is twice differentiable at x = 0 (D) none

Ans. B

Sol. $$f'(x) = \underset{h\to 0}{Lim}\ \frac{f(x+h)-f(x)}{h} = \underset{h\to 0}{Lim}\ \frac{f(h)+|x|h+xh^2}{h}$$

Q f (0) = 0; hence $f'(x) = \underset{h\to 0}{Lim}\left(\frac{f(h)-f(0)}{h}+|x|+xh\right)$

$f'(x) = f'(0) + |x| = |x|$

Q12. For x > 0, let $h(x) = \begin{cases} \frac{1}{q} & if\ x = \frac{p}{q} \\ 0 & if\, x\, is\, irrational \end{cases}$ where p & q > 0 are relatively primeintegers

then which one does not hold good?

(A) h(x) is discontinuous for all x in $(0, \infty)$

(B) h(x) is continuous for each irrational in $(0, \infty)$

(C) h(x) is discontinuous for each rational in $(0, \infty)$

(D) h(x) is not derivable for all x in $(0, \infty)$

Ans. A

Sol. Let $x = \frac{2}{3}$ which is rational

$\Rightarrow h\left(\frac{2}{3}\right) = \frac{1}{3}$

$\lim\limits_{t\to 0} h\left(\frac{2}{3}+t\right) = 0 \Rightarrow$ discontinuous at $x \in Q$

Let $x = \sqrt{2} \notin Q$ consider $\sqrt{2} = 1.41401235839$

$h\left(\sqrt{2}\right) = h\left(\frac{1414023583}{10^{10}}\right) = \frac{1}{10^{10}} \to 0$

Hence h is continuous for all irrational $\Rightarrow$ A.

Q13. Let f (x) = max.$\left\{\left|x^2-2|x|\right|, |x|\right\}$ and g (x) = min. $\left\{\left|x^2-2|x|\right|, |x|\right\}$ then

(A) both f (x) and g (x) are non differentiable at 5 points.

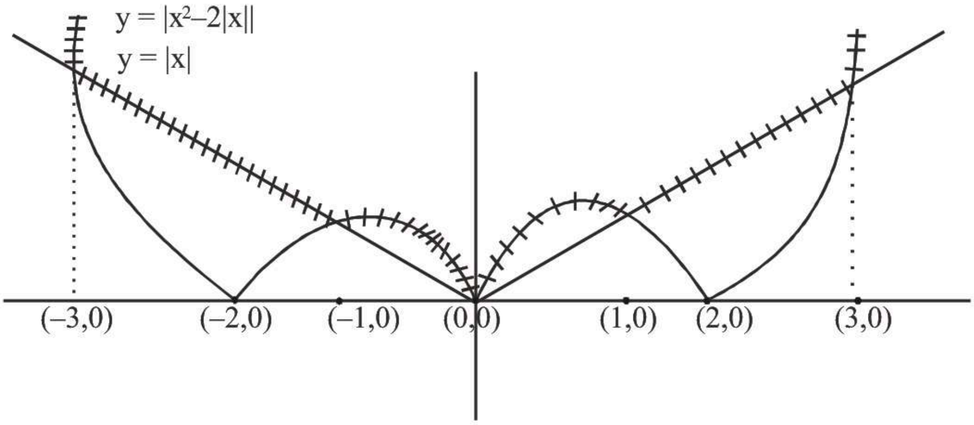
(B) both f (x) and g (x) are non differentiable at 5 points.

(C) number of points of non differentibility for f (x) and g (x) are 7 and 5 respectively.

(D) both f (x) and g (x) are non differentiable at 3 and 5 points respectively.

Ans. B

Sol. $f(x) = \max\{|x^2 - 2|x||, |x|\}$



Clearly f(x) is not differentiable at x = –3, –1, 0, 1, 3

Q14. If f is defined on an interval [a, b]. Which of the following statement(s) is/are INCORRECT?

(A) If f (a) and f (b), have opposite sign, then there must be a point $c \in (a, b)$ such that f (c) = 0.

(B) If f is continuous on [a, b], f (a) < 0 and f (b) > 0, then there must be a point $c \in (a, b)$ such that f (c) = 0

(C) If f is continuous on [a, b] and there is a point c in (a, b) such that f (c) = 0, then f (a) and f (b) have opposite sign.

(D) If f has no zeroes on [a, b], then f (a) and f (b) have the same sign.

Ans. D, C, A

Sol. (A) Wrong because function must be continuous -

(B) Correcrt by IMVT

(C) This may not necessary

(D) If function is discontinuous, then it may be wrong.

Q15. If f(x) = Min (tan x, cot x) then:

(A) f(x) is discontinuous at $x = 0, \frac{\pi}{4}$ & $\frac{5\pi}{4}$ (B) f(x) is continuous at $x = \frac{\pi}{2}$ & $\frac{3\pi}{2}$

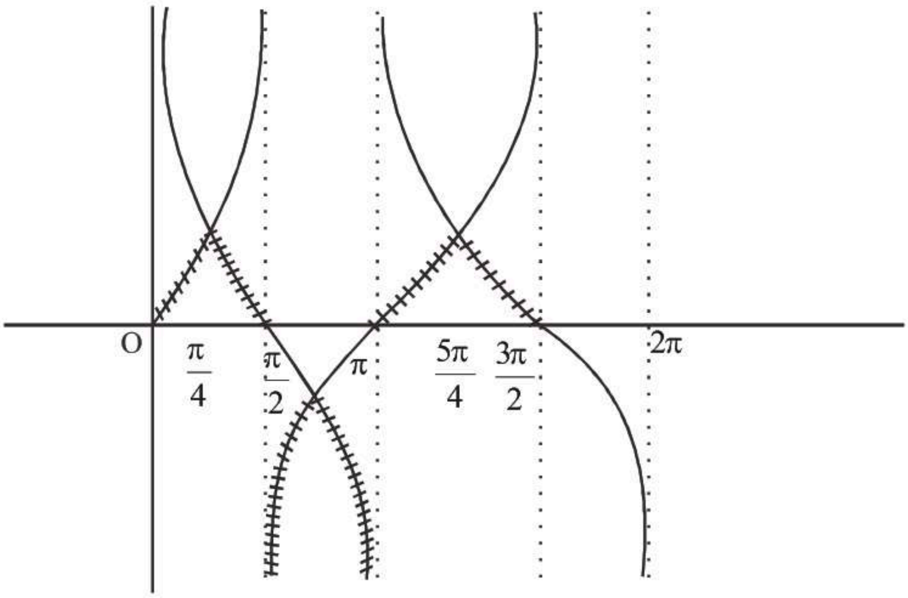
(C) $\int_0^{\pi/2} f(x)\,dx = 2\ ln\sqrt{2}$ (D) f(x) is periodic with period π.

Ans. C, D

Sol. f(x) = Min(Tanx, cotx)

y = tan x

y = cot x



obviouly (A) and (B) are wrong

$$\int_0^{\frac{\pi}{2}} f(x)dx$$

$$\int_0^{\frac{\pi}{4}} \tan x dx + \int_0^{\frac{\pi}{2}} \cot x dx$$

$$= \ln\sec x\Big]_0^{\frac{\pi}{4}} + \ln\sin x\Big]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= (\ln\sqrt{2} - 0) + (\ln\sin\frac{\pi}{2} - \ln\frac{1}{\sqrt{2}})$$

$$= (\ln\sqrt{2})$$

Clearly period of f(x) is π .

Q16. If f (x) = |2x + 1| + |2x - π| + tan x then f (x) is

(A) Continuous at all points except $x = n\pi + \frac{\pi}{2}$ (where $n \in I$)

(B) Continuous at $x = -\frac{1}{2}$, but not differentiable at $x = \pi/2$

(C) Not differentiable at $x = n\pi + \frac{\pi}{2}$(where $n \in I$) (D) Limit at $x = \frac{\pi}{2}$does not exist.

Ans. A, B, C, D

Sol. $f(x) = |2x + 1| + |2x - \pi| + \tan x$

By observation of given function it is clear options (A), (B), (C) & (D) are correct.

Q17. The function defined as $f(x) = \lim_{n\to\infty} \begin{bmatrix} \cos^{2n} x & if & x < 0 \\ \sqrt[n]{1+x^n} & if & 0 \le x \le 1 \\ \frac{1}{1+x^n} & if & x > 1 \end{bmatrix}$. Which of the following does not hold good?

(A) continuous at $x = 0$ but discontinuous at $x = 1$.

(B) continuous at $x = 1$ but discontinuous at $x = 0$.

(C) continuous both at $x = 1$ and $x = 0$.

(D) discontinuous both at $x = 1$ and $x = 0$.

Ans. A, B, C

Sol. $f(0^-) = 0; \quad f(0^+) = 1; \quad f(0) = 1 \quad \Rightarrow$ discontinuous at $x = 0$

$f(1^-) = 1; \quad f(1) = 1; \quad f(1^+) = 0 \quad \Rightarrow$ discontinuous at $x = 1$

Q18. f is a continous function in [a, b]; g is a continuous functin in [b, c] A function h (x) is defined as

$h(x) = f(x)$ for $x \in [a, b)$
$= g(x)$ for $x \in (b, c]$
if $f(b) = g(b)$, then

(A) h(x) has a removable discontinuity at x=b.

(B) h(x) may or may not be continuous in [a, c]

(C) $h(b^-) = g(b^+)$ and $h(b^+) = f(b^-)$

(D) $h(b^+) = g(b^-)$ and $h(b^-) = f(b^+)$

Ans. A, C

Sol. Given f is continuous in [a, b] ....(1)

g is continuous in [b, c] ....(2)

$f(b) = g(b)$ ....(3)

$$\left.\begin{aligned} h(x) &= f(x) && \text{for } x \in [a, b) \\ &= f(b) = g(b) && \text{for } x = b \\ &= g(x) && \text{for } x \in (b, c] \end{aligned}\right\} \quad ....(4)$$

h (x) is continuous in $[a, b) \cup (b, c]$ [using (1), (2)]

also $f(b–) = f(b)$; $g(b^+) = g(b)$ ....(5) [using (1), (2)]

$\therefore h(b^-) = f(b^-) = f(b) = g(b) = g(b^+) = h(b^+)$ [using (4), (5)]

now, verify each alternative. Of course! $g(b^-)$ and $f(b^+)$ are undefined.

$h(b^-) = f(b^-) = f(b) = g(b) = g(b^+)$ and $h(b^+) = g(b^+) = g(b) = f(b) = f(b^-)$

hence $h(b^-) = h(b^+) = f(b) = g(b)$ and h (b) is not defined $\Rightarrow$ (A).

Q19. Let f be a continuous function on the closed interval [a, b]. Then which of the following statement(s)
need not be true?

(A) There is a number c in the open interval (a, b) such that $f(a) < f(c) < f(b)$.

(B) There is a number c in the closed interval [a, b] such that $f(c) \geq f(x)$ for all x in [a, b].

(C) There is a number c in the open interval (a, b) such that $f'(c) = 0$

(D) There is a number c in the open interval (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Ans. A, C, D

Sol. (A) Can be true only if function is strictly increasing in (a, b).

(B) Correct because by extreme value theorem for continuous function.

(C) False. Why $f'(c) = 0$?

(D) Can be true if differentiable in (a, b)].

Q20. Let $f(x) = \begin{cases} x^2 \cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\frac{1}{x}, & x > 0 \end{cases}$,

then which of the following is(are) correct?

(A) f(x) is continuous but not differentiable at x = 0.

(B) f(x) is continuous and differentiable at x = 0.

(C) f '(x) is continuous but not differentiable at x = 0.

(D) f' (x) is discontinuous at x = 0.

Ans. D, B

Sol. Clearly f(x) is continuous and differentiable at x = 0 such that $f'(0) = 0$

$$f'(x) = \begin{cases} \sin\frac{1}{x} + 2x\cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ 2x\sin\frac{1}{x} - \cos\frac{1}{x}, & x > 0 \end{cases}$$

$\therefore$ Clearly f '(x) is discontinuous at x = 0.

Q21. If f (x) = min. $(1, x^2, x^3)$, then

(A) f (x) is continuous $\forall\, x \in R$ (B) f' (x) > 0, $\forall\, x > 1$

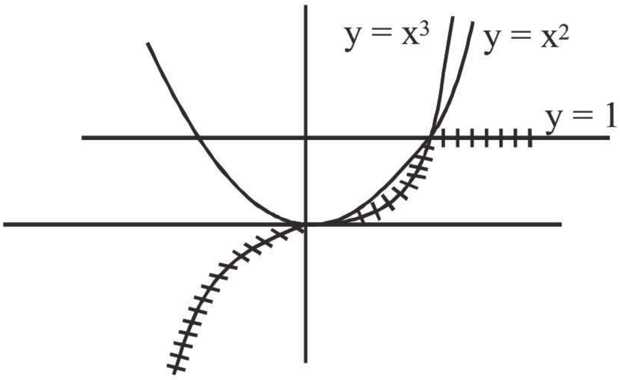
(C) f(x) is not differentiable but continuous $\forall\, x \in R$

(D) f(x) is not differentiable for two values of x

Ans. C, A

Sol. $f(x) = \text{Min}(1, x^2, x^3)$

$y = 1$
$y = x^2$
$y = x^3$



$$f(x) = \begin{cases} x^3 & ; \ x \in (-\infty, 1] \\ 1 & ; \ x \in (1, \infty) \end{cases}$$

$$f'(x) = \begin{cases} 3x^2 & ; \ x \in (-\infty, 1] \\ 0 & ; \ x \in (1, \infty) \end{cases}$$

hence correct options are (A) & (C).

Q22. $f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ |||x|-1|-1| & ; \quad x \ge 1 \end{cases}$, then

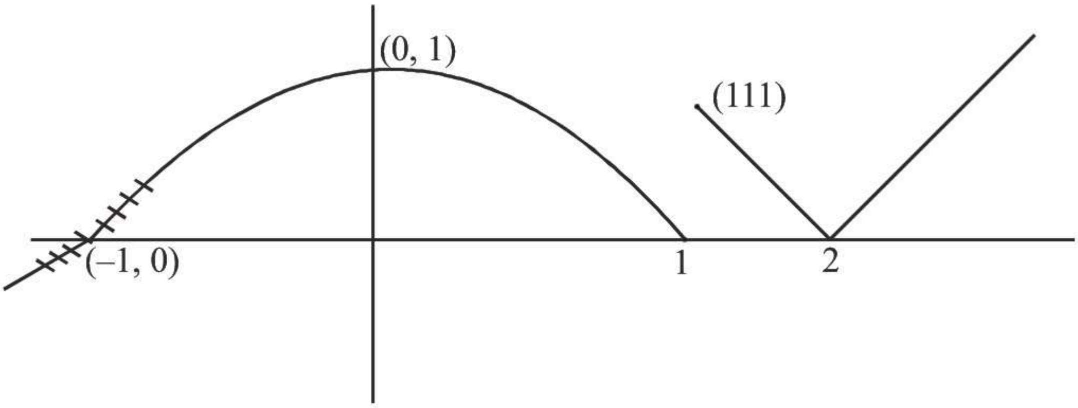
(A) f(x) is non differentiable at exactly three points (B) f(x) is continuous in $(-\infty, 1]$

(C) f(x) is differentiable in $(-\infty, -1)$

(D) f(x) is finite type of discontinuity at x = 1, but continuous at x = –1

Ans. D, C, A

Sol. By Graph



(A) at x = –1, 1, 2 function is not differentiable.

(B) Correct

(C) Correct

(D) Correct

Q23. If f(x) = |||x|–2|+p| have more than 3 points of non-derivability then the value of p can be

(A) 0 (B) –1 (C) –2 (D) 2

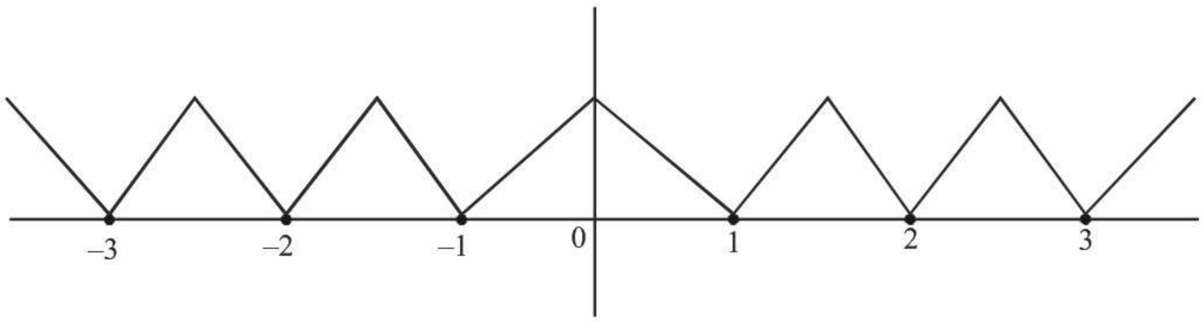
Ans. B, C

Sol. $y = \Big|||x|-2|+p\Big|$

By option (B) & (C)

p = –1

$y = \Big|||x|-2|-1\Big|$



Clearly it has more than '3' points of non-derivability.